# Two Physics Defects in the MoCHII Line-Cooling Fits: Discovery by a Fixed-State Cooling Budget, Repair, and a Comparison with the Other Monte Carlo Codes


**Abstract**

MoCHII reproduced the ionization structure of the Lexington/Meudon H II-region benchmarks well but ran hotter than every published code: $\langle T[N_p N_e]\rangle$ = 9159 K at HII40 against a published range of 7720–8199 K, and 7370 K at HII20 against 6402–6749 K. Candidate causes were excluded one at a time by direct experiment, which left the cooling function itself. A *fixed-state* comparison — evaluating MoCHII's cooling expressions on Cloudy c25.00's own converged zone state, so that no feedback can hide a difference — localized the problem immediately: on an identical state the Tier-1 fits that drive the thermal balance delivered 0.72 of Cloudy's total cooling while the Tier-2 $n$-level solve that produces the line output delivered 0.98. The gap was two defects in the offline fitting pipeline, both in `chianti_cooling.py` and its caller. D1: the lower-level populations were distributed among the ground-term fine-structure levels by Boltzmann factors, which at $kT \gg \Delta E_{\rm fs}$ is the statistical-weight (high-density) limit and suppressed the far-infrared fine-structure cooling ($O$ III $\times 1.64$, $N$ III $\times 2.9$, $C$ II $\times 2.5$ too small); volume-integrated impact 23.9% of the total cooling. D2: the excitation energy was taken from the `.scups` Burgess–Tully *scaling* energy rather than the observed `.elvlc` level difference, and entered both the Boltzmann factor and the radiated photon ($O$ II: 30661 versus 26811 cm$^{-1}$, a factor 0.583 at 8000 K); impact 10.6%. Missing channels were not the story ($S$ IV, the largest, is 1.2%). Rebuilding all 28 fits from the $n_e \to 0$ limit of statistical equilibrium with observed transition energies moved HII40 to **7894 K, inside the published range**, and HII20 to 6376 K, 0.4% below the published floor; $L$(H$\beta$), $R_{\rm out}$ and the line ratios all moved into or next to the published spread at the same time. That zero-density limit still overcooled the low-critical-density lines at nebular densities, leaving both benchmarks at the cool edge of the range; suppressing the line cooling to the local $n_e$ with a $(T, n_e)$ table built from the same $n$-level solve — now the default, `par%cooling_model = local_ne` — removes the far-infrared overestimate, brings the fixed-state line total from 1.116× to 1.01× Cloudy's, and superseded the two zero-density temperatures at once by 8211 and 7025 K, matching Cloudy c25.00 (8210, 6910 K); the zero-density fits are retained as the `low_density` option. Corrections to the diffuse field made after that carried the production figures to **8169 K** at HII40 — inside the published 7720–8199 K range and 0.50% below Cloudy's 8210 K — and **6925 K** at HII20, which sits above the published 6402–6749 K range but within 0.10% of Cloudy c25.00's 6910 K, itself above that range: both current-generation codes now run hotter than the 1995-era HII20 spread, and they agree with each other. A comparison of the cooling inventories of the other Monte Carlo photoionization codes is included; it turned up defects on their side as well, and one omission common to all five codes.


## 1. Context: what was left after everything else was excluded

The Lexington/Meudon H II-region benchmarks HII40 (40 kK blackbody) and HII20 (20 kK blackbody) are the standard multi-code comparison for photoionization codes; the published columns used here are those of Wood et al. (2004, their Table 1), which agree entry by entry with the compilation of Ercolano et al. (2003, their Tables 4 and 5). Nine codes participate (GF, HN, DP, TK, PH, RS, RR, BE and WME in the notation of that table), so every "published range" quoted in this memo is the span of those nine columns. The CLOUDY c25.00 column, C25, is a run made for this work and is deliberately *not* counted in it: a 2025 code is not one of the original participants, and keeping it separate is what makes it usable as an independent current-generation reference. MoCHII matched the published ionization structure but sat above every participating code in temperature:

| | MoCHII (before) | published range | CLOUDY c25.00 |
|---|---|---|---|
| HII40 $\langle T[N_p N_e]\rangle$ / K | 9159 | 7720–8199 | 8210 |
| HII20 $\langle T[N_p N_e]\rangle$ / K | 7370 | 6402–6749 | 6910 |

that is +14.1% over the HII40 published median (8026 K) and +11.6% over a current CLOUDY run of the same deck. The following candidate causes were each excluded by a direct experiment rather than by argument:

- **Charge exchange.** A stage-mapping error was found and corrected (transition $i$ must use $\mathtt{chex}(Z,i)$, not $(Z,i+1)$); the corrected rates changed HII40 $T_e$ by −0.7%, not by 9%.
- **Convergence of the reference.** The stored MOCASSIN baseline was underconverged; a full-quality re-run moved *it* by −230 K and left the disagreement in place.
- **Spatial resolution.** A $25^3$ MOCASSIN grid gives 8195 K against 8176 K on $13^3$.
- **Recombination-coefficient vintage.** Replacing the Badnell 2023 metal RR+DR by MOCASSIN's Pequignot 1991 / Nussbaumer–Storey 1983 set inside MoCHII moved $T_e$ by +0.2% — and in the wrong direction.
- **O III collision strengths.** The dominant coolant's ${}^3P \to {}^1D$ collision strength agrees between CHIANTI v7 (the MOCASSIN vintage) and v11 to 0.9% (2.283 versus 2.262 at $10^4$ K), worth $< 30$ K.
- **Ionizing-band discretization.** Aligning the band edges on the ionization thresholds fixed a real He and metal over-ionization but left $T_e$ unchanged.
- **Transport treatment.** Switching from case-B on-the-spot to the explicit Monte Carlo diffuse field *raises* $T_e$ by +2.6% (HII40) and +1.6% (HII20), so it cannot be the cause of an excess.
- **Trace-species coupling.** Adding the metal electrons to $n_e$ and the metal photoheating to the balance moved HII40 from 9159 to 9178 K, +0.2%.
- **Photoheating.** The mean excess energy per photoionization is $\langle h\nu - 13.6\,\mathrm{eV}\rangle_{\mathrm{H}} = 3.9$ eV and $\langle h\nu - 24.6\,\mathrm{eV}\rangle_{\mathrm{He}} =$ 4.1 eV, ordinary values for a mildly hardened 40 kK field, and the volumetric heating equals $n_e n_p \alpha \langle h\nu - h\nu_{\mathrm{th}}\rangle$ with a standard $\alpha$.

A second apparent discrepancy turned out to be a consequence of the first rather than an independent one. $L(\mathrm{H}\beta)$ was 12% low at HII40. Photon counting only holds for a radiation-bounded model: the temperature excess lowers $\alpha_{\mathrm{B}}$ by $\times 0.897$, grows the Strömgren volume by $\times 1.115$, and drives the ionization front past the prescribed outer radius, so part of the ionizing luminosity leaves the model. The predicted leaked fraction, 10.3%, matched the measured one (recombinations $3.788 \times 10^{49}$ against $Q_{\mathrm{H}} = 4.265 \times 10^{49}$ s$^{-1}$, an 11.2% escape), and HII20, which stays bounded, showed no such deficit. So one open item remained, on the cooling side.

## 2. The fixed-state cooling-budget method

Comparing two converged models confounds three things: the thermal balance, the ionization structure, and the radiation field. Each feeds back on the others, and a cooling deficit hides itself by raising the temperature until the books balance again. The diagnosis therefore used a *fixed-state* comparison, in which no feedback is possible:

1. run CLOUDY c25.00 on the HII40 benchmark and save its zone-by-zone cooling inventory, heating inventory, overview state and ionization structure;
2. read the state $(T_e, n_e, n_H, x_{ion})$ of every zone;
3. evaluate MoCHII's own cooling expressions on that state, channel by channel, and volume-integrate both sides on $4\pi r^2\, dr$.

Any difference is then attributable to the atomic data and the cooling formulas alone. The CLOUDY run is the physics block of the distributed test-suite deck `tsuite/auto/hii_paris.in`, unmodified, with only output commands appended (`tests/cloudy_c25/hii40_cool.in`); grains and cosmic rays are absent from that deck, as its own log states.
Two evaluations of the MoCHII side are carried through, because MoCHII computes metal line cooling twice, by two independent routes:

**Tier-1** the fitted curves $\Lambda(T) = T^{-1/2} \sum_i A_i \exp(-T_i/T)$ read from `data/atomic/cooling_<ion>.txt`. These drive the thermal balance and are the object of this memo.

**Tier-2** the $n$-level statistical-equilibrium solve on the `nlevel_<ion>.txt` models, used at output time for line luminosities.

Both are built from the same CHIANTI v11.0.2 files by the same offline pipeline, so a disagreement between them on an identical state is internal evidence, independent of CLOUDY.

**Consistency checks performed before the comparison was believed.**

- Parsed ion fractions reproduce the CLOUDY file values exactly (O III 0.4570305, S IV 0.0982921 in zone 1).
- The sum of the ion cooling channels of an element reproduces that element's `.cooleach` column to 0.9886–0.9999; the residual is the free-bound share that `.cooleach` also carries.
- Free-free: the `.cool` "FF" entry equals the `.cooleach` $FF_H + FF_M$ sum to 1.0000.
- The `.ovr` O III column agrees with the `.ele_o` $O^{+2}$ column to $4.4 \times 10^{-3}$.
- CLOUDY's own heating and cooling totals balance to 0.99987, and its heating is 4.574 eV per ionizing photon, split 90.8% H I, 8.5% He I — the same physical picture as MoCHII's.
- The Python Tier-2 solver used here reproduces MoCHII's own Fortran line luminosities on the same state to 0.1% for all ten benchmark ions, so it is a faithful stand-in for the compiled code.

**Result.** Table 1 is the comparison. The reading is unambiguous: with the same state, the same database and the same code base, the Tier-2 route reproduces CLOUDY's total cooling to 2% while the Tier-1 fits that actually set the temperature deliver 72% of it. The defect is therefore in the construction of the Tier-1 fits, not in the atomic database, not in the missing channels, and not in the ionization structure.

Table 1: HII40 volume-integrated cooling on the CLOUDY c25.00 state. "Tier-1" is MoCHII's fitted cooling as the thermal balance used it before the fix; "Tier-2" replaces the metal line cooling by MoCHII's $n$-level solve on the same state. Both are ratios to the CLOUDY column. The free-bound rows are quoted case A, which is why they exceed unity; CLOUDY reports free-bound cooling net of escaping photons, so the matched bookkeeping is the case-B line at the foot of the table. Both MoCHII columns are as they stood at the time of the diagnosis, when S IV had neither a cooling fit nor an $n$-level model; it has both now (Section 4).

| Channel | CLOUDY [erg s$^{-1}$] | % of $L_{\rm cool}$ | Tier-1 | Tier-2 |
|---|---|---|---|---|
| [$O$ III] lines | $1.0020 \times 10^{38}$ | 32.06 | 0.587 | 1.024 |
| [$O$ II] lines | $4.8892 \times 10^{37}$ | 15.65 | 0.633 | 0.980 |
| [$S$ III] lines | $4.7559 \times 10^{37}$ | 15.22 | 0.703 | 1.000 |
| H recombination (fb) | $3.6809 \times 10^{37}$ | 11.78 | 1.635 | 1.635 |
| free-free (net) | $2.3947 \times 10^{37}$ | 7.66 | 0.998 | 0.998 |
| [$N$ II] lines | $1.4818 \times 10^{37}$ | 4.74 | 0.922 | 0.990 |
| [$S$ II] lines | $6.7404 \times 10^{36}$ | 2.16 | 1.008 | 0.998 |
| [$Ne$ III] lines | $6.0443 \times 10^{36}$ | 1.93 | 0.933 | 1.299 |
| [$N$ III] lines | $5.9069 \times 10^{36}$ | 1.89 | 0.433 | 1.019 |
| [$C$ II] lines | $5.5241 \times 10^{36}$ | 1.77 | 1.383 | 0.940 |
| [$Ne$ II] lines | $4.1043 \times 10^{36}$ | 1.31 | 0.694 | 0.997 |
| [$S$ IV] lines | $3.8916 \times 10^{36}$ | 1.25 | 0.000 | 0.000 |
| He I recombination (fb) | $3.4075 \times 10^{36}$ | 1.09 | 1.720 | 1.720 |
| H I collisional lines | $2.2490 \times 10^{36}$ | 0.72 | 0.989 | 0.989 |
| C III] lines | $1.2741 \times 10^{36}$ | 0.41 | 0.822 | 0.990 |
| [$O$ I] lines | $3.1112 \times 10^{35}$ | 0.10 | 0.841 | 0.837 |
| He I collisional lines | $1.9430 \times 10^{35}$ | 0.06 | 0.000 | 0.000 |
| metal recombination (fb) | $1.6610 \times 10^{35}$ | 0.05 | 0.000 | 0.000 |
| **Total (mapped)** | $3.1244 \times 10^{38}$ | 99.98 | **0.819** | **1.077** |
| **Total, case-B recombination** | | | **0.723** | **0.981** |

Two further readings of Table 1 are worth recording. First, the ions whose Tier-1 ratio is farthest below unity ($N$ III 0.433, $O$ III 0.587, $O$ II 0.633, $Ne$ II 0.694, $S$ III 0.703) are exactly the ions whose Tier-2 ratio is near unity, so the two tiers disagree with each other and only one of them can be right. Second, the missing channels are small: [$S$ IV] 10.5 $\mu$m, the largest one absent from MoCHII altogether, is 1.245% of the budget, [$O$ IV] is 0.007%, and every other omission is below 0.2%. Adding channels could not have closed a 28% gap.

# 3. The two defects

## 3.1 D1: ground-term populations were a high-density model

The fitting library built the cooling as $\Lambda = \sum_{l,u} f_l\, q_{lu}(T)\, \Delta E_{lu}$ and set the lower-level fractions $f_l$ by Boltzmann factors among the levels of the ground term: `cooling_effective` of `tools/fitting/chianti_cooling.py` called with the population model `pop='ground_term'`, from `tools/fitting/fit_cooling.py` (the call site is line 246 of that file, now reading `cooling_low_density_limit`).

For a ground term whose fine-structure splittings are far below $kT$ this is wrong in a specific way. At $T = 8000$ K, $kT = 5560$ cm$^{-1}$, while the $O$ III $^3P$ levels lie at 0, 113 and 306 cm$^{-1}$: every Boltzmann factor is within a few percent of unity, so the population collapses onto the statistical weights, $g = 1, 3, 5$, and the lowest level receives 0.115. That is the LTE limit within the ground term, which is reached only for $n_e \gg n_{crit}$ of the fine structure. The true population at the nebular $n_e = 110$ cm$^{-3}$, where $n_e \ll n_{crit}$, is the opposite limit: the metastable fine-structure levels are drained radiatively faster than they are collisionally populated, and the lowest level holds $\sim 0.75$. Table 2 lists both for the benchmark ions.

The mechanism by which this loses cooling is the far-infrared fine-structure lines. The optical lines of $O$ III are excited out of all three $^3P$ levels, so their total is nearly insensitive to how the $^3P$ population is distributed (the fractions sum to one either way). The [$O$ III] 52 and 88 $\mu$m lines, however, are excitations *within* the term, out of the lowest levels into the higher ones; emptying the lowest level empties them. At the benchmark state the $n$-level solve puts 46.7% of the $O$ III cooling in 52 + 88 $\mu$m and 53.3% in the optical lines — and the defective Tier-1 fit returned 0.594 of the Tier-2 value, that is, almost exactly the optical share alone.

Volume-integrated over the HII40 model, D1 accounts for 23.9% of the total cooling.

Table 2: D1: the ground-term population model. "Boltzmann" is the fraction in the lowest level under the population model that was used; "true" is the fraction from the $n$-level statistical equilibrium at $n_e = 110$ cm$^{-3}$. The last column is the old Tier-1 fit divided by the Tier-2 solve at the same state; it exceeds unity for $C$ II because the [$C$ II] 158 $\mu$m line is already collisionally saturated at this density (Section 6). $T = 8000$ K.

| Ion | ground-term levels ($g, E$ [cm$^{-1}$]) | Boltzmann $f_1$ | true $f_1$ | Tier-1/Tier-2 |
|---|---|---|---|---|
| $O$ III | (1, 0) (3, 113) (5, 306) | 0.115 | 0.752 | 0.594 |
| $S$ III | (1, 0) (3, 298) (5, 833) | 0.123 | 0.907 | 0.700 |
| $N$ II | (1, 0) (3, 49) (5, 131) | 0.113 | 0.280 | 0.905 |
| $N$ III | (2, 0) (4, 174) | 0.340 | 0.885 | 0.425 |
| $C$ II | (2, 0) (4, 63) | 0.336 | 0.414 | 1.933 |
| $Ne$ II | (4, 0) (2, 780) | 0.697 | 1.000 | 0.698 |
| $Ne$ III | (5, 0) (3, 643) (1, 920) | 0.587 | 0.999 | 0.731 |
| $O$ II | (4, 0) (4, 26831) (6, 26811) | 0.980 | 1.000 | 0.596 |
| $S$ II | (4, 0) (4, 14853) (6, 14885) | 0.853 | 0.996 | 1.017 |
| $O$ I | (5, 0) (3, 158) (1, 227) | 0.563 | 0.999 | 1.002 |

## 3.2 D2: the Burgess–Tully scaling energy was used as a physical energy

The excitation energy of every transition was taken from the `.scups` header field, `de_erg = tr["de"] * RY_ERG`. That field is the Burgess & Tully *scaling* energy: a theoretical value whose only role is to build the scaled temperature on which the collision strength is tabulated. It is not the physical transition energy, and it entered the calculation twice — in the Boltzmann factor $\exp(-\Delta E/kT)$ of the excitation rate and as the energy of the radiated photon.

The physical requirement is that both be the observed `.elvlc` level difference. Detailed balance ties the collisional excitation rate to the same level gap that appears in the statistical-equilibrium matrix; if the cooling sum and the $n$-level solve use different gaps for the same transition, they are describing different atoms. For [$O$ II] 3726/3729 the `.scups` value is 30661 cm$^{-1}$ against an observed 26811 cm$^{-1}$: the ratio of Boltzmann factors alone is $\exp(-3850\,\mathrm{cm}^{-1}\,hc/kT) = 0.50$ at 8000 K, and with the smaller radiated energy the net factor is 0.583.

Volume-integrated over HII40, D2 accounts for 10.6% of the total cooling.

Table 3: Isolating the two defects. $\Lambda$ at $T = 8000$ K [erg cm$^3$ s$^{-1}$] per $n_e n_{ion}$, with each defect repaired separately and then together. "both" must reproduce the $n_e \to 0$ limit of the $n$-level solve, and does. The last column is the $n$-level solve at the nebular $n_e = 110$ cm$^{-3}$, which additionally carries the density suppression the fits do not (Section 6).

| | $\Lambda$(8000 K) | | | | $n$-level solve | |
|---|---|---|---|---|---|---|
| Ion | as fitted | D2 fixed | D1 fixed | both | $n_e \to 0$ | $n_e = 110$ |
| $O$ III | 2.691e–21 | 3.328e–21 | 4.401e–21 | 4.980e–21 | 4.970e–21 | 4.564e–21 |
| $O$ II | 8.465e–22 | 1.452e–21 | 8.465e–22 | 1.452e–21 | 1.452e–21 | 1.421e–21 |
| $S$ III | 3.179e–20 | 3.179e–20 | 4.691e–20 | 4.690e–20 | 4.691e–20 | 4.543e–20 |
| $N$ II | 4.988e–21 | 5.655e–21 | 5.470e–21 | 6.155e–21 | 6.108e–21 | 5.512e–21 |
| $N$ III | 7.412e–22 | 7.251e–22 | 2.140e–21 | 2.109e–21 | 2.105e–21 | 1.742e–21 |
| $Ne$ II | 7.019e–22 | 7.022e–22 | 1.007e–21 | 1.007e–21 | 1.007e–21 | 1.007e–21 |
| $Ne$ III | 2.143e–21 | 2.143e–21 | 2.932e–21 | 2.933e–21 | 2.932e–21 | 2.929e–21 |
| $C$ II | 5.892e–22 | 5.891e–22 | 1.450e–21 | 1.450e–21 | 1.366e–21 | 3.048e–22 |
| $S$ II | 3.458e–20 | 3.470e–20 | 3.458e–20 | 3.470e–20 | 3.468e–20 | 3.396e–20 |
| $O$ I | 4.702e–22 | 4.702e–22 | 4.693e–22 | 4.693e–22 | 4.691e–22 | 4.691e–22 |
| $C$ III | 4.724e–23 | 5.887e–23 | 4.724e–23 | 5.887e–23 | 5.845e–23 | 5.844e–23 |

## 3.3 Temperature closure

The two defects together are worth 34.1% of the HII40 cooling. The net shortfall of the fits against the $n$-level solve is smaller, 25.8%, because the low-density limit gives back –8.3% through the density suppression it does not carry.

Whether that deficit explains the temperature excess can be checked without running the code, because at fixed state the cooling is a smooth power of $T$. The logarithmic slope of the total MoCHII cooling at the Cloudy state is $d\ln\Lambda/d\ln T = 1.748$. Scaling $T_e$ by a factor $f$ until MoCHII's cooling equals Cloudy's — Cloudy balances its own heating at $f = 1$, and the two codes' photoheating agrees — gives

$$f = 1.111 \longrightarrow T_{\rm e} = 9118\ {\rm K},$$

against the 9159 K MoCHII actually converged to and the 8210 K of CLOUDY. Repeating with case-B recombination bookkeeping gives 9544 K, so the estimate brackets the observed value. In other words the cooling defect accounts for essentially the entire +11.6% excess.

A second, independent estimate uses MoCHII's own converged state: there, the Tier-2 cooling is 1.270× the Tier-1 cooling at the same temperature, and equals it at $f = 0.856$, that is $\langle T_{\rm e}\rangle = 7836$ K in place of 9159 K. The two estimates agree that repairing the fits moves the benchmark from above the published range to inside it.

# 4. The fix

## 4.1 The right population model

`cooling_low_density_limit()` (`tools/fitting/chianti_cooling.py`, line 232) replaces the ground-term Boltzmann model with the $n_{\rm e} \to 0$ limit of the $n$-level statistical equilibrium,

$$\Lambda(T) = \sum_u q_{1u}(T)\,\Delta E_{1u},$$

the sum running over every `.scups` transition out of the lowest level. The physical argument is the one the defect got backwards: at $n_{\rm e} \ll n_{\rm crit}$ every ion sits in its lowest level, so collisional excitation happens out of that level only, and each excitation is followed by a radiative cascade that ultimately radiates the full $\Delta E_{1u}$ whatever route it takes down. The total cooling is therefore the excitation power out of the ground level, which is exactly what the $n$-level solve returns as $n_{\rm e} \to 0$. That identity is the strongest verification available and is checked directly (Section 4.4).

## 4.2 The right energies

A new function, `transition_energy_erg()` at line 213 of the same file, returns the observed `.elvlc` level difference. `cooling_effective` now calls it at line 333, in place of the old `de_erg = tr["de"] * RY_ERG`. Transitions whose observed energies do not order the two levels return zero and are skipped rather than silently contributing a negative energy. The `.scups de` field is retained for its only legitimate use, building the scaled temperature.

## 4.3 A fit ladder derived from the atomic structure

The repair also removed a piece of hand tuning that had been hiding the defect. In the low-density limit the cooling is

$$\Lambda(T) = T^{-1/2}\sum_u \left[\tfrac{C}{g_1}\,\Upsilon_{1u}(T)\,\Delta E_{1u}\right] e^{-T_u/T}, \qquad T_u = \Delta E_{1u}/k,$$

which *is* the fitted form $\Lambda = T^{-1/2}\sum_i A_i e^{-T_i/T}$ apart from the slow temperature dependence of $\Upsilon$. The fitted $T_i$ are therefore excitation temperatures, not free parameters, and `ground_excitation_ladder()` (`fit_cooling.py`, line 115) now builds the starting ladder by clustering the transitions in $\log T_u$ with the share of the cooling each carries as the weight. The hand-tuned guess lists that the previous version carried for each ion are gone, which is why the same script can now be pointed at a new ion without tuning.

## 4.4 Verification of the rebuilt fits

Three independent checks were run on all 28 rebuilt files (Table 4):

1. **Fit against its own target.** The fitted curve against the directly evaluated $n_e \to 0$ statistical-equilibrium cooling over $10^3$–$10^5$ K. The H/C/N/O/Ne/S set is within 0.7% over the whole range except *C* III (5.5% at the cold end, 0.93% at $10^4$ K) and *S* II (2.2%); the depleted Si, Cl and Ca ions reach 4–28% at the ends of the range, which is accepted because at abundances $\sim 3 \times 10^{-7}$ their share of the thermal balance is $\sim 10^{-4}$ and their diagnostics run through the exact Tier-2 solve.

2. **Fit against a different code path.** The fitted curve against the Tier-2 $n$-level solve at $n_e = 1$ cm$^{-3}$ — a different data file and a different solver. Agreement is 0.1–0.2% for the benchmark ions. The outliers are informative rather than alarming: Fe II 0.873, Fe III 0.842 and Ca V 1.045 differ because their $n$-level models are level-truncated, while the Tier-1 fit sums the full `.scups` list.

3. **New against old.** The ion-by-ion multiplier at 5, 8 and 10 kK, which is the quantitative statement of how much cooling the defects were losing.

Table 4: The rebuilt Tier-1 fits. "max err" and "err at $10^4$ K" are against the directly evaluated $n_e \to 0$ statistical-equilibrium cooling over $10^3$–$10^5$ K; "new/old" is the rebuilt fit divided by the defective one; the last column is the independent Tier-2 solve at $n_e = 1$ cm$^{-3}$ divided by the new fit. *S* IV is new, so it has no old file. The Ar, Mg, Fe, Si, Cl and Ca ions are omitted here; their multipliers run 0.66–2.83 and their default abundances are zero.

| | fit quality | | new/old | | | Tier-2($n_e$=1) |
|---|---|---|---|---|---|---|
| Ion | max | at $10^4$ K | 5 kK | 8 kK | 10 kK | / new |
| H I | 0.52% | 0.20% | 1.012 | 1.008 | 1.006 | 1.000 |
| C I | 0.27% | 0.02% | 1.189 | 1.049 | 1.017 | — |
| *C* II | 0.35% | 0.12% | 2.956 | 2.460 | 1.742 | 0.943 |
| *C* III | 5.48% | 0.93% | 1.442 | 1.246 | 1.191 | 0.971 |
| *N* I | 0.58% | 0.14% | 1.161 | 1.092 | 1.069 | — |
| *N* II | 0.66% | 0.31% | 1.557 | 1.233 | 1.163 | 0.993 |
| *N* III | 0.70% | 0.23% | 2.877 | 2.845 | 2.564 | 0.998 |
| *O* I | 0.20% | 0.08% | 1.035 | 0.998 | 0.997 | 1.000 |
| *O* II | 0.27% | 0.07% | 2.616 | 1.716 | 1.492 | 1.000 |
| *O* III | 0.32% | 0.07% | 2.731 | 1.836 | 1.501 | 0.999 |
| *Ne* II | 0.40% | 0.12% | 1.397 | 1.432 | 1.446 | 1.001 |
| *Ne* III | 0.34% | 0.11% | 1.501 | 1.369 | 1.247 | 1.001 |
| *S* II | 2.23% | 0.96% | 1.005 | 1.003 | 1.003 | 1.001 |
| *S* III | 0.23% | 0.01% | 1.944 | 1.474 | 1.349 | 1.001 |
| *S* IV | 0.61% | 0.16% | | (new ion) | | 1.002 |

### 4.5 S IV added

[*S* IV] 10.51 $\mu$m was the one channel of any size that MoCHII did not carry at all. S IV was already tracked in the ionization cascade, so adding it was the intended data operation: `cooling_s_4.txt` (three exponential terms, max fit error 0.61%) and `nlevel_s_4.txt` from the same pipeline. In the rebuilt HII40 model [*S* IV] 10.51 $\mu$m comes out at 0.1868 of H$\beta$ against 0.1894 from the Cloudy c25.00 emissivity file, a 1.4% difference.

All 28 coefficient files carry updated provenance headers stating the CHIANTI version, the population model, the energy convention, the fit range and the maximum fit error, together with the density caveat of Section 6.

## 5. Benchmarks after the fix

Table 5 is the before-and-after on the two benchmark models. HII40 moves from 9159 K, above every published code, to 7894 K, inside the published range. HII20 moves to 6376 K, 0.4% below the published floor. These are the zero-density-limit (`low_density`) temperatures, which the density-suppressed default of Section 6 superseded one day later; they are kept in the table because they are what the repair of the two defects by itself is worth, and the audit trail of this memo is the point. The *current* figures, in the last column of the table, are 8169 K at HII40 and 6925 K at HII20, and they change the HII20 verdict:

| | published (nine codes) | Cloudy c25.00 | MoCHII | verdict |
|---|---|---|---|---|
| HII40 | 7720–8199 | 8210 | **8169** | inside published; 0.50% below Cloudy |
| HII20 | 6402–6749 | 6910 | **6925** | above published; 0.22% above Cloudy |

At HII20 it is not MoCHII alone that exceeds the published range: Cloudy c25.00 exceeds it too, by 2.4%, and MoCHII agrees with that current-generation reference to 0.10%. The published HII20 range is a 1995-era spread of nine codes, and both codes written since sit above it. The earlier reading of this section — "0.4% below the published floor" — belonged to the zero-density fits and inverted the sign of the present comparison; it is retained above only as the historical value.

The photon budget closes at the same time, which is the confirming test of the leakage argument of Section 1. With the cooler gas the ionization front no longer runs past the prescribed outer radius: the recombination rate rises from 0.888 to 0.953 of $Q_{\rm H}$ at HII40, the mean $x_{\rm H^+}$ in the outermost 2% of the ionized volume falls to 0.07, and $R_{\rm out}$ becomes predictable at both benchmarks ($1.43 \times 10^{19}$ and $8.69 \times 10^{18}$ cm against published 1.45–$1.47 \times 10^{19}$ and 8.70–$9.01 \times 10^{18}$). $L$(H$\beta$) follows to within 3% of the published values.

The line ratios move with the temperature. Table 6 lists the principal coolants; the systematic excess is gone, and the residual pattern is what one expects from a model sitting at the cool edge of the range: the temperature-sensitive optical lines of the doubly ionized species come out low ([*S* III] 9532+9069 is the clearest case), while the infrared fine-structure lines, which are nearly temperature-independent, sit inside the published spread.

Table 5: The Lexington/Meudon benchmarks before and after the cooling fix, and where they stand now. “published” is the range over the nine codes of the published compilation, which does not include C25; C25 is CLOUDY c25.00 run on its own unmodified test-suite deck. The $n_e \to 0$ column is the state directly after the fit repair — the zero-density (`low_density`) cooling, threshold-aligned 32 ionizing bins, the explicit Monte Carlo diffuse field with the He I channels, case-B recombination-line tables. The “current” column is the production default: the density-suppressed `local_ne` cooling of Section 6, continuous stellar photon energies, Milne ground continua and the resolved He I diffuse cascade. It first raised $\langle T \rangle$ to 8211 and 7025 K and then settled at 8169 and 6925 K; Section 6 lists the intermediate steps. The recombination fraction was a diagnostic of the photon-budget argument and was not re-measured for the current default.

| | before | $n_e \to 0$ | current | C25 | published |
|---|---|---|---|---|---|
| **HII40** | | | | | |
| $L(\mathrm{H}\beta)/10^{37}$ erg s$^{-1}$ | — | 1.95 | 1.94 | 2.06 | 2.01–2.10 |
| $\langle T[N_p N_e] \rangle$ / K | 9159 | 7894 | **8169** | 8210 | 7720–8199 |
| $T_{\mathrm{inner}}$ / K | — | 7270 | 7535 | 7812 | 7370–8318 |
| $R_{\mathrm{out}}/10^{19}$ cm | not predicted | 1.43 | 1.44 | 1.47 | 1.45–1.47 |
| $\langle \mathrm{He}^+ \rangle / \langle \mathrm{H}^+ \rangle$ | 0.667 | 0.715 | 0.706 | 0.741 | 0.715–0.829 |
| recombinations / $Q_{\mathrm{H}}$ | 0.888 | 0.953 | — | — | — |
| **HII20** | | | | | |
| $L(\mathrm{H}\beta)/10^{36}$ erg s$^{-1}$ | — | 4.70 | 4.70 | 4.99 | 4.83–5.04 |
| $\langle T[N_p N_e] \rangle$ / K | 7370 | 6376 | **6925** | 6910 | 6402–6749 |
| $T_{\mathrm{inner}}$ / K | — | 6752 | 6936 | 7155 | 6562–7224 |
| $R_{\mathrm{out}}/10^{18}$ cm | — | 8.69 | 8.89 | 8.97 | 8.70–9.01 |
| $\langle \mathrm{He}^+ \rangle / \langle \mathrm{H}^+ \rangle$ | 0.047 | 0.047 | 0.046 | 0.043 | 0.034–0.077 |
| recombinations / $Q_{\mathrm{H}}$ | — | 0.971 | — | — | — |

## 6. Density suppression of the line cooling at the local $n_e$

The repaired fits are the $n_e \to 0$ limit, so they contain no collisional de-excitation. Lines whose critical density lies at or below the nebular density are therefore overestimated. On the same fixed CLOUDY state the rebuilt Tier-1 line total is $1.116\times$ CLOUDY’s while the Tier-2 solve at the local $n_e$ gives $1.011\times$; the difference is the density suppression, and it is concentrated in exactly the expected places:

Table 6: HII40 line intensities relative to H$\beta$ after the fix. “published” is the range over the nine codes of the published compilation, which does not include C25. [$O$ III] 5007+4959 was 4.28 before the fix. Like the $n_e \to 0$ column of Table 5, these are the zero-density-limit values; the density-suppressed default warms the model and raises the temperature-sensitive optical lines.

| Line | MoCHII | C25 | published |
|---|---|---|---|
| [$O$ III] 5007+4959 | 2.06 | 2.46 | 1.64–3.27 |
| [$O$ III] 52+88 $\mu$m | 2.41 | 2.40 | 2.10–2.42 |
| [$O$ III] 4363 | 0.0033 | 0.0046 | 0.0022–0.0070 |
| [$O$ II] 3726+3729 | 1.90 | 2.32 | 1.60–2.26 |
| [$N$ II] 6584+6548 | 0.626 | 0.669 | 0.669–0.852 |
| [$N$ III] 57.3 $\mu$m | 0.280 | 0.286 | 0.223–0.311 |
| [$S$ III] 9532+9069 | 0.91 | 1.00 | 1.13–1.44 |
| [$S$ III] 18.7 $\mu$m | 0.494 | 0.504 | 0.535–0.750 |
| [$S$ II] 6716+6731 | 0.299 | 0.304 | 0.130–0.315 |
| [$Ne$ II] 12.8 $\mu$m | 0.202 | 0.200 | 0.177–0.217 |
| [$Ne$ III] 15.5 $\mu$m | 0.278 | 0.216 | 0.263–0.429 |
| $C$ III] 1907+1909 | 0.029 | 0.062 | 0.041–0.091 |
| [$S$ IV] 10.51 $\mu$m | 0.187 | 0.189 | — |

| Channel | Tier-1 / CLOUDY | Tier-2 / CLOUDY | excess, points of $L_{\rm line}$ |
|---|---|---|---|
| [$C$ II] 158 $\mu$m etc. | 2.763 | 0.940 | +3.93 |
| [$O$ III] (52 + 88 $\mu$m share) | 1.126 | 1.024 | +5.12 |
| [$Ne$ III] | 1.300 | 1.299 | +0.73 |
| [$N$ III] | 1.236 | 1.019 | +0.56 |
| [$N$ II] | 1.095 | 0.990 | +0.57 |
| line total | 1.116 | 1.011 | |

[$C$ II] 158 $\mu$m has $n_{\rm crit} \sim 50$ cm$^{-3}$, so at $n_e = 110$ the low-density limit overestimates it by 4.5× (Table 3, $1.366 \times 10^{-21}$ against $3.048 \times 10^{-22}$). The [$O$ III] far-infrared pair is the same effect at a higher $n_{\rm crit}$. [$Ne$ III] is *not* a density effect — Tier-1 and Tier-2 agree with each other and both exceed CLOUDY by 30% — so that row is a collision-data difference between CHIANTI v11 and CLOUDY’s Ne III data, and is noted here as a separate open item.

Run with the zero-density fits used directly, the consequence for the benchmarks was a slight cooling bias at $n_{\rm H} = 100$: both models sat at the cool edge of the published range, and HII20 fell 0.4% below its floor. MoCHII removes it by suppressing the line cooling to the local density. A $(T, n_e)$ table, built once at setup (`nlevel_cooling_mod`) by solving the same $n$-level atom on a grid ($\log_{10} T = 2$–6, $\log_{10} n_e = -6$–10), stores for each ion the suppression ratio $S_{\rm sup}(T, n_e) = \Lambda_{\rm SE}(T, n_e)/\Lambda_{\rm SE}(T, n_e \to 0)$ and the low-density reference; the runtime coefficient is $\Lambda = \Lambda_{\rm fit}\,[\,f_{\rm cov} S_{\rm sup} + (1 - f_{\rm cov})\,]$ with $f_{\rm cov} = \min(1,\ \Lambda_{\rm SE}(T, n_e \to 0)/\Lambda_{\rm fit})$ the fraction of the low-density cooling the (often truncated) $n$-level model resolves, so a truncated model cannot bias the total and $n_e \to 0$ returns the fit exactly. Because the table is built from the very solve that produces the diagnostic lines, the thermal balance and the emergent line list are now one physics. This is the default (`par%cooling_model = local_ne`); the pure zero-density fits are kept as the `low_density` option, exact in diffuse gas and needed in any case for the planetary-nebula densities where the suppression is much larger.

On the fixed CLOUDY state the default delivers the Tier-2 column of the table above, 1.011× CLOUDY's line total in place of 1.116×, and the converged benchmarks rose to **8211** K at HII40 — matching CLOUDY c25.00's 8210 K to 0.7 K — and 7025 K at HII20 (+1.7% against CLOUDY's 6910 K), in place of the zero-density 7894 and 6376 K. This confirms the fixed-state prediction quantitatively: removing the far-infrared overestimate — the difference between the 1.116× and 1.011× totals — is what carries the converged temperature the remaining distance to the current-generation reference. Ne III remains the one channel still 30% above CLOUDY, a collision-data difference rather than a density effect.

**From 8211/7025 to the current 8169/6925 K.** The two temperatures above are those measured when `local_ne` became the default; they are not the current ones, and the difference is not in the cooling. Three later changes to the stellar and diffuse radiation field moved them, each recorded with its own gate:

| Change | Plan document in `docs/` | HII40 | HII20 |
|---|---|---|---|
| zero-density fits (`low_density`) | this memo | 7894 | 6376 |
| `local_ne` default | `COOLING_LOCAL_NE_PLAN.md` | 8210 | 7024 |
| C I/N I $n$-level, energy order | (same) | 8211 | 7025 |
| continuous stellar photon energies | `CONTINUOUS_ENERGY_PHOTON_PLAN.md` | 8194 | 6916 |
| Milne ground continua | `MILNE_DIFFUSE_LYC_IMPLEMENTATION_PLAN.md` | 8202 | 6922 |
| resolved He I diffuse cascade | `HEI_CASCADE_DIFFUSE_PLAN.md`, `HEI_584_CONVERSION_PLAN.md` | **8169** | **6925** |

Sampling the stellar spectrum at continuous photon energies, rather than at band-averaged energies, removed the discretization bias in the photoheating and in the threshold cross sections. Sampling the three ground recombination continua from the Milne free-bound spectrum $E^2\sigma_{\rm pi}(E)\exp[-(E-E_{\rm th})/kT]$ replaced an exponential approximation, and the shift is above the Monte Carlo noise: the seed-to-seed spread over four scrambled-Sobol ensembles is 0.002% median on the line ratios against a 0.40% median Milne shift. Resolving the He I diffuse cascade replaced three constants — statistical branch weights, a flat $2^1S$ two-photon spectrum, and the 584 Å photon treated as ionizing with probability one — by effective recombination coefficients and the Drake–Victor–Dalgarno $2^1S$ spectrum, which lowers the effective ionizing content of the helium cascade and so lowers HII40 by 40 K. None of this touches the cooling function, and none of it disturbs the conclusion of this memo: HII40 stays inside the published range throughout, and HII20 stays above it — within 1.7% of CLOUDY c25.00 at the `local_ne` default and within 0.2% from the continuous-energy step onward.

**The H-collider term is still low-density (needs re-baselining).** One line-cooling term is not yet density-suppressed: the H-impact [$C$ II] 158 $\mu$m / [$O$ I] 63 $\mu$m fine-structure cooling of the photodissociation-region package (`metal_cooling_H`), a two-level low-density formula. It is weighted by $x_{\rm HI}$, so it is ~0 in the ionized benchmark zones (the H II gates are unaffected), but it needs the same saturation in dense PDR gas ($n_{\rm H} \sim 5 \times 10^3$ in the `isrf_cloud` example) — a separate follow-up. The photodissociation-region test `tests/pdr/pdr_L5.in` is otherwise unchanged: the PDR shell has $n_e = A_C n_H$ plus 6% from S, Mg and Fe, $T_e = 258$ K, $C$ II = 1.000, and there $n_e \sim 2 \times 10^{-2}$ cm$^{-3}$ is far below $n_{crit}$ so the low-density limit is the correct limit. The stored PDR gate references and the figures and examples built on the earlier cooling still need refreshing to the density-suppressed default, and `tests/g2a_thermal/check_g2a.py` still refers to a coefficient file under its former name.

# 7. Cooling inventories of the other Monte Carlo codes

The temperature of a photoionized nebula is set almost entirely by what is in the cooling inventory, and Sections 2–3 showed how a defect there hides itself behind the thermal balance. The same audit was therefore turned on the four Monte Carlo photoionization codes closest to MoCHII in method: the code of Wood et al. (2004), which is the WME column of the benchmark tables; CMacIonize, its open successor, which inherits Wood's ion ordering by its own account; Mocassin; and Torus, whose photoionization loop borrows from both Wood's code and Mocassin. This is an internal record, so everything found is written down, including defects on their side. Line numbers below were read from the sources in `~/RT_Codes/`.

## 7.1 Wood et al. `ionize`

**Ion set.** Ten five-level ions are read from `atom4.dat` in the order N I, $N$ II, $O$ I, $O$ II, $O$ III, $Ne$ III, $S$ II, $S$ III, $C$ II, $C$ III (`setcool.f`:25, `DO 100 I=1,10`; `linecool1.f`:74), plus two two-level ions, $N$ III and $Ne$ II (`linecool1.f`:160–171, abundances assigned at `ioneng.f`:168–171). The code is self-consistent, but two of its own comments are not: the header list at `linecool1.f`:4–5 puts carbon in slots 11–12 rather than 9–10, and the loop comment `!omit 9,10 = C cooling` on line 74 is stale — those slots are in the loop. `lines.f`:366–367 carries the corrected ordering.

**Chlorine and argon data are present but unread.** `atom4.dat` contains fourteen five-line blocks; blocks 11–14 are Cl II, Ar III, Cl III and Ar IV (`atom4.dat`:51, 56, 61, 66). They are never read, both because the read loop stops at ten and because the arrays are dimensioned `(10,10)` and `(10,5)` (`setcool.f`:7–8). Chlorine and argon cooling are therefore absent although their data sit in the file.

**Defect: N I and O I line cooling.** `linecool1.f`:42–64 evaluates hardcoded collision-strength fits into `OM(1,*)` and `OM(3,*)` for N I and O I; then `linecool1.f`:81–83, *inside* the ion loop, overwrites the whole row,

```
DO MM=1,10 ; OM(J,MM)=CS(J,MM)*T4**CSE(J,MM) ; END DO
```

so lines 42–64 are dead code. The replacement values taken from `atom4.dat` are zero: the O I collision strengths and exponents (`atom4.dat`:13–14) are all zero, and the N I row (`atom4.dat`:3) is zero except entries 5 and 10, which in the index order of `setcool.f`:14 are the $(2, 3)$ and $(4, 5)$ transitions — both between excited levels. Every ground-state excitation is zero, the level system becomes homogeneous, and both ions cool exactly zero. Reproducing `linecool1.f`:85–123 with the file values at $T = 8000$ K, $n_e = 100$ cm$^{-3}$ gives $N$ I and $O$ I cooling = 0 against $4.75 \times 10^{-19}$ erg s$^{-1}$ for $O$ III on the same inputs.

There is a qualification that makes this worse rather than better. `cs` is one array shared by the cooling routine and the diagnostic routine, and `lines.f`:406–428 writes its own N I/O I fits into `CS` rather than `OM`; because the corresponding

`CSE` entries are zero, those values survive the overwrite at `linecool1.f`:81–83. The diagnostic call sits inside the iteration loop (`mcionize.f`:575–577) while the thermal balance starts only at iteration 5 (`itbal.f`:27), so by the time `linecool` runs, `CS(1,*)` and `CS(3,*)` hold values left from the last cell of the previous iteration's diagnostic pass. N I and O I cooling is thus not identically zero at run time, but frozen at one unrelated temperature, with no $T$ rescaling. Either way it is not the physical term.

**Defect: the cooling and diagnostic paths use different atomic data.** The two two-level ions carry different collision strengths in the two routines:

| | cooling (`linecool1.f`) | diagnostic (`lines.f`) | CHIANTI v11 |
|---|---|---|---|
| $N$ III 57.3 $\mu$m $\Upsilon$ | 0.701 $T_4^{0.136}$ (:31, :164) | 1.45 (:394, :540) | 1.378 |
| $Ne$ II 12.8 $\mu$m $\Upsilon$ | 0.368 (:32, :169) | 0.303 (:395, :545) | 0.314 |

at $10^4$ K. Only the cooling path carries the $T_4^{0.136}$ factor. This one can be adjudicated, because both codes describe the same transition: Wood's $N$ III $A$-value, $4.77 \times 10^{-5}$ s$^{-1}$, is the CHIANTI v11 value to three figures. Evaluating the CHIANTI v11.0.2 collision strength through MoCHII's own descaling gives $\Upsilon = 1.378$ at $10^4$ K and 1.296 at 8000 K, so the *diagnostic* value 1.45 is right to 5% and the *cooling* value is a factor 2.1 too small. [$N$ III] 57.3 $\mu$m is 1.89% of the HII40 cooling budget and lies below its critical density there, so the cooling is low by nearly that factor in a channel of that size — while the code's own line output for the same transition uses the larger value. For $Ne$ II, CHIANTI gives 0.314, again closer to the diagnostic value; the cooling value is 17% high. The hardcoded N I blocks are also assigned to different transition indices in the two files (`linecool1.f`:44–50 versus `lines.f`:408–414, with indices 3/4, 6/7 and 8/9 interchanged); which ordering is intended cannot be settled from the sources, since neither cites the row order of the table it transcribes.

**S IV slot never filled.** The variable `csiv10` is declared and passed through the argument lists (`lines.f`:46, 74, 204, 363, 388) and used at `lines.f`:250 to fill `em(:,:,:,28)`, but no line of the code ever assigns it. There is no S IV cooling anywhere in the code either.

**O III $^1S_0$ level energy.** `atom4.dat`:21 gives the $O$ III levels as $0, 113.4, 306.8, 20271, 43072$ cm$^{-1}$. The observed $^1S_0$ energy is 43185.7 cm$^{-1}$ (CHIANTI `o_3.elvlc`; CMacIonize carries 43186 at `LineCoolingData.cpp`:550), so the file value is low by 113.7 cm$^{-1}$, which is numerically the $^3P_1$ energy — a transcription slip. It raises the ground-to-$^1S_0$ Boltzmann factor by 2.1% at 8000 K and shifts the nominal [$O$ III] 4363 wavelength to 4385.8 Å, so the line and any temperature diagnosed from it carry that bias. The cooling itself barely notices, because the $^1S_0$ channel is small.

**H and He terms.** The whole budget is `loss = Lrec + Lff + Lc` (`ioneng.f`:191). `RECcool` (`glfunc.f`:15–29) has exactly two terms, $H^+$ and $He^+$; $He^{++}$ is not tracked at all (`h0find.f` solves only $h^0$ and $he^0$), so there is no He III recombination cooling and free-free is $Z = 1$ only. There is no H I collisional-excitation (Ly$\alpha$) cooling and no collisional-ionization cooling.

### 7.2 CMacIonize

**Ion set: thirteen ions, S IV included.** Ten five-level ions in Wood's order (`LineCoolingData.hpp`:38–61 — the file says outright that "the order is historical: this was the order of the elements in Kenny's code") plus three two-level ions, *N* III, *Ne* II and *S* IV (:68–77). The S IV data are real, not a placeholder: Martin, Zalubas & Musgrove (1990) energies, a Pradhan (1995) $A$-value of $7.73 \times 10^{-3}$ s$^{-1}$, and fits to Saraph & Storey (1999) (`LineCoolingData.cpp`:1341–1376). So CMacIonize carries the one channel Wood omits entirely and MoCHII was missing.

**Atomic data provenance.** The header (`LineCoolingData.hpp`:126–133) states that many of the sources belong to the IRON Project and were located through its online database, with the source list at :136–182 (Berrington 1988; Froese Fischer & Tachiev 2004; Galavis, Mendoza & Zeippen 1997, 1998; Kisielius et al. 2009; Lennon & Burke 1994; Tayal 2000, 2008; Tayal & Zatsarinny 2010; Zatsarinny & Tayal 2003). The compilation is a mixture, not exclusively IRON Project.

**N I and O I are complete.** Seven-parameter fits for all ten transitions of each ion (`LineCoolingData.cpp`:112–118 for N I, from Tayal 2000; :353–359 for O I, from Zatsarinny & Tayal 2003 and Berrington 1988), fed abundances at `TemperatureCalculator.cpp`:374–390. CMacIonize does not carry Wood's N I/O I defect, nor the *O* III $^1S_0$ transcription slip.

**The He$^+$ difference is in the cooling, not the recombination rate.** The He$^+$ recombination *coefficient* is identical in the two codes: the same Verner & Ferland (1996) HeIa fit with the same constants $3.294 \times 10^{-11}$, 15.54, 0.309, $3.676 \times 10^7$, 1.691 (`VernerRecombinationRates.cpp`:174–183 against Wood `recomb.f`:17–19), so their ratio is unity at every temperature. What differs by about 11% is the He$^+$ recombination *cooling* coefficient: Wood uses $L_{\mathrm{He}^+} = 2.6 n_e n_{\mathrm{He}^+} T^{0.32}$ (`glfunc.f`:26) while CMacIonize uses $1.55 n_e n_{\mathrm{He}^+} T^{0.3647}$ (`TemperatureCalculator.cpp`:489) — which is exactly the expression Wood carries commented out on the following line, `glfunc.f`:27. Both are attributed to Black (1981), Table 3. The ratio Wood/CMacIonize is 1.146, 1.123, 1.111 and 1.077 at 5, 8, 10 and 20 kK.

**Budget.** The code's own comment at `TemperatureCalculator.cpp`:347–351 lists three terms and no more: line cooling, free-free, and H and He recombination. There is no H I collisional-excitation cooling and no collisional-ionization cooling, and He$^{++}$ is absent from the code altogether (`ElementNames.hpp` defines only `ION_H_n` and `ION_He_n`), so free-free is $Z = 1$ only and there is no He III recombination cooling.

### 7.3 Mocassin

**Budget and solve.** `coolInt = coolFF + coolRec + coolColl` (`update_mod.f90`:1251), with two dust terms added under `lgDust .and. lgPhotoelectric` (:1253–1256). `thermBalance` is called from `iterateT` (:392), which re-solves the ionization balance at every trial temperature (:365) and then steps on the residual `heatInt - coolInt` (:397) — the same arrangement as MoCHII's bisection with the ionization solved inside it. Collisionally excited line photons are not transported: when a diffuse packet is drawn as a line photon it escapes (`photon_mod.f90`:925–927), so the line cooling is optically thin, as in MoCHII. Only the resonance lines of a user list are followed, and that is for dust absorption downstream of the cooling (`nResLines`, `common_mod.f90`:533).

**Ion set: whatever the data directory supplies.** The heavy-element sum runs `elem = 3, nElements` with `nElements = 30` (`constants_mod.f90`:54) and `ion = 1, min(elem+1, nstages)`, `nstages` defaulting to 7 and capped at 10 (`set_input_mod.f90`:82, :176–186). An ion enters when its element has a nonzero abundance and its file exists: `makeElements` (`hydro_mod.f90`:478) walks the 259 slots of `data/fileNames.dat` (:511–518) and sets `lgDataAvailable` from the result (:701–731). 116 of the 259 files are present. For the benchmark composition (He, C, N, O, Ne, S) that is 26 ions at HII40's default `nstages` — C I–IV, N I–V, O I–VI, Ne II–VII and S II–VI — and 25 at the `nstages 6` of the HII20 deck (`benchmarks/gas/HII20/input.in`:22). Adding an ion is a data operation here too. The coverage reaches higher ionization stages than MoCHII's registry, which for these five elements stops at C III, N III, O III, Ne III and S IV; that is immaterial at 40 kK (Cloudy puts 0.007% of the budget in [*O* IV]) but not for the hard-spectrum models on MoCHII's plan.

**Level populations are a statistical-equilibrium solve at the local $n_e$.** `equilibrium` (`emission_mod.f90`:1913–2122) interpolates each $\Upsilon$ by a cubic spline in $\log_{10} T$ (:2015–2032), forms $q_{lu} = 8.63 \times 10^{-6}\, \Upsilon\, e^{-\Delta E/kT}/(g_l\sqrt{T})$ from the observed level energies of the file (:2040–2047), builds the rate matrix with $n_e q$ in both directions plus the $A$-values (:2063–2089), solves it by LU decomposition and normalizes the populations (:2091–2101), and returns $A_{ji}\Delta E_{ji} n_j$ for every transition with a nonzero $A$ (:2104–2116). Collisional de-excitation is therefore carried, and the far-infrared fine-structure lines saturate above their critical density. MoCHII now does the same (Section 6): where its $n_e \to 0$ fits deliver 1.116× Cloudy's line cooling on the fixed Cloudy state, the $(T, n_e)$ table built from its own $n$-level solve delivers the suppressed 1.011× — a difference that is a factor 4.5 on [*C* II] 158 $\mu$m alone — and that suppressed value is what the thermal balance uses by default. Both codes therefore carry the density suppression in the thermal balance.
The recombination-pumping term of the same routine is dormant. It requires a trailing `br, z, a_r(4), a_d(5)` block in the ion file (`hydro_mod.f90`:651–664, consumed at `emission_mod.f90`:1992–2004 and :2087); none of the 116 files carries one, so `br = 0` and the populations are purely collisional — the same assumption as MoCHII's Tier-2 models. Cooling and line output come from the same routine and the same files. The driver is duplicated, once inside the thermal balance (`update_mod.f90`:1264–1319, called at :1100) and once for the emissivities (`emission_mod.f90`:838–904), and the two differ only in the unit factor ($1.9865 \times 10^{-16}$ against $1.9865 \times 10^{9}$). Mocassin therefore cannot carry the kind of cooling/diagnostic data split found in Wood's code.

**Two truncations.** `nForLevels = 17` (`constants_mod.f90`:55) dimensions `forbiddenLines` (`grid_mod.f90`:89) and the line output uses all 17 levels, but the cooling sum runs `j,k = 1..10` only (`update_mod.f90`:1105–1106). Thirty of the 116 ion models have more than ten levels, so for those the cooling and the line list are not built from the same level set. At nebular temperatures this costs nothing: the *O* III levels that fall outside the sum start at 142382 cm$^{-1}$, a Boltzmann factor of $8 \times 10^{-12}$ at 8000 K. Iron is the second case. Its 142-level model (`nForLevelsLarge`, :56) is written to a separate array, `forbiddenLinesLarge` (`update_mod.f90`:1278–1293), which the `coolColl` sum never reads, so Fe II line cooling is absent from the thermal balance although its lines are output. Both truncations are latent: every composition shipped with the code has Fe = 0, and an iron- or calcium-bearing model would in any case stop at `emission_mod.f90`:1974–1978, since Fe VI (19 levels) and Ca VII (27) exceed `nForLevels` and are reached at `nstages` $\geq 6$ and $\geq 7$.

**Atomic data: CHIANTI v7, with two pre-CHIANTI files among the main coolants.** Every ion file states its provenance in its own header. `data/oiii.dat`:2 reads "Atomic data from version 7 of the CHIANTI database", and so do all the benchmark files except two. `data/siii.dat`:2–3 gives "Upsilons from Mendoza (1983, Proc. IAU Symp. 103, p. 143); Aij's from Mendoza & Zeippen (MNRAS, 199, 1025, 1982)" on five levels and four temperatures (5000–20000 K), and `data/ci.dat`:2–3 gives Pequignot & Aldrovandi (1976), with $A$-values from Nussbaumer & Rusca (1979), on six levels and six temperatures (500–20000 K). [$S$ III] is 15.2% of the HII40 cooling budget (Table 1), so MOCASSIN's second-largest metal coolant runs on a 1983 compilation sampled at four temperatures.

Table 7 puts a number on what the vintage is worth. For each ion the $n_e \to 0$ line cooling was evaluated twice with the same formula and the same upper levels, once from the MOCASSIN file and once from CHIANTI v11.0.2 through MoCHII's own descaling. The procedure reproduces the $O$ III cross-check quoted in Section 1: the total $\Upsilon(^3P \to {}^1D)$ is 2.283 from the file against 2.262 from v11, a ratio of 1.009. The reading of the table is that the older data are systematically *higher* on three of the four largest metal coolants — $S$ III 1.223 (its $\Upsilon(^3P \to {}^1D)$ is 8.39 against 6.96), $N$ II 1.199, $S$ II 1.163, with $N$ III 1.154 and $O$ II 1.088 — while $O$ III, $Ne$ III, $S$ IV and $O$ I agree to 3%. $C$ III] is 1.457 on a 0.41% channel.

Two entries fall the other way, and the larger one is worth stating carefully. N I is a factor 17 low against v11, and this is a genuine change of calculation rather than a transcription slip: MOCASSIN's file gives $\Upsilon(^4S \to {}^2D_{5/2})$ = 0.027 at $10^4$ K and cites Tayal (2000), CMACIONIZE's independent fit to the same Tayal (2000) data gives 0.025 (`LineCoolingData.cpp`:112–118 through the fit form at :1590–1598), and CHIANTI v11, which uses Tayal (2006), gives 0.337. The two calculations converge above $10^5$ K (0.550 against 0.563). So $N$ I cooling and the [$N$ I] 5200 diagnostic differ by an order of magnitude between the 2000 and the 2006 calculation, in exactly the partly neutral gas where N I lives. $C$ I is 0.640, the 1976 compilation against v11.

**H I.** Ly$\alpha$ and Ly$\beta$ collisional excitation from the Mathis coefficients (`update_mod.f90`:1072–1089). The energy radiated per excitation is taken as one Rydberg for both lines (`hcRyd`, :48) rather than the 10.20 and 12.09 eV of the transitions, so the coefficients are effective rather than literal — and the product is right. Against MoCHII's CHIANTI v11 H I low-density fit (`data/atomic/cooling_h_1.txt`) the expression gives $2.211 \times 10^{-25}$ against $2.276 \times 10^{-25}$ at 8000 K and $4.140 \times 10^{-24}$ against $4.223 \times 10^{-24}$ at $10^4$ K, that is 0.97–0.98, falling to 0.94–0.95 at 6 and 20 kK; with the literal line energies it would be 25% low, because the two-term form has to stand in for the whole Lyman series. The hard cut `coolColl = 0` below 5000 K (:1079) removes nothing measurable. There is no separate two-photon or free-bound cooling term; free-bound sits inside `coolRec`.

**He.** There is no He I or He II collisional-excitation cooling: the collisionally excited line sum starts at `elem = 3` in both drivers (`update_mod.f90`:1103, :1272; `emission_mod.f90`:846). $He^{++}$ free-free and recombination cooling are present (:1014–1035), the Hummer (1994) Table 1 fits with the polynomial doubled and evaluated at $\log_{10}(T/4)$, and $He^+$ free-free has its own fit (:1047–1048, attributed to Hummer & Storey 1998, Table 6). The $He^+$ *recombination* cooling, however, is the $H^+$ polynomial copied coefficient for coefficient: compare :1059–1062 with :989–991, all five

Table 7: MOCASSIN's ion models for the benchmark coolants: levels and tabulated temperatures in `data/<ion>.dat`, the source of the effective collision strengths as the file's own header gives it, and $\Lambda_{\rm M}/\Lambda_{11}$, the ratio of the $n_{\rm e} \to 0$ line cooling computed from that file to the same quantity computed from CHIANTI v11.0.2 through MOCHII's descaling, at $T$ = 8000 K with the same upper levels on both sides. All files are labeled CHIANTI v7 except `ci.dat` and `siii.dat`, whose headers are quoted in the text. The temperature grid is the CHIANTI-wide 0.2-dex grid from 100 K to $2.51 \times 10^6$ K where it has 23 points.

| Ion | file | levels | $T$ pts | collision strengths | $\Lambda_{\rm M}/\Lambda_{11}$ |
|---|---|---|---|---|---|
| C I | `ci.dat` | 6 | 6 | Pequignot & Aldrovandi (1976) | 0.640 |
| C II | `cii.dat` | 10 | 23 | Blum & Pradhan (1992) | 0.966 |
| C III | `ciii.dat` | 10 | 23 | Berrington et al. (1985, 1989) | 1.457 |
| N I | `ni.dat` | 13 | 23 | Tayal (2000) | 0.058 |
| N II | `nii.dat` | 15 | 23 | Stafford et al. (1994) | 1.199 |
| N III | `niii.dat` | 10 | 23 | Stafford, Bell & Hibbert (1994) | 1.154 |
| O I | `oi.dat` | 7 | 23 | Zatsarinny & Tayal (2003) | 1.004 |
| O II | `oii.dat` | 13 | 23 | Pradhan et al. (2006) | 1.088 |
| O III | `oiii.dat` | 15 | 23 | Burke & Lennon (1994) | 1.028 |
| Ne II | `neii.dat` | 2 | 23 | Saraph & Tully (1994) | 0.905 |
| Ne III | `neiii.dat` | 9 | 23 | McLaughlin & Bell (2000) | 0.997 |
| S II | `sii.dat` | 5 | 23 | Ramsbottom, Bell & Stafford (1996) | 1.163 |
| S III | `siii.dat` | 5 | 4 | Mendoza (1983) | 1.223 |
| S IV | `siv.dat` | 12 | 23 | Tayal (2000) | 0.992 |

constants identical, while the comment above it cites the helium source. At $10^4$ K it gives $2.40 \times 10^{-25}$ erg cm$^3$ s$^{-1}$ against $3.88 \times 10^{-25}$ for MOCHII's $k_{\rm B}T\alpha_{\rm B}$(He II), $4.95 \times 10^{-25}$ for Wood's Black-1981 coefficient and $4.46 \times 10^{-25}$ for CMACIONIZE's, so it is low by a factor 1.6–2.1 against all three. Only part of that is the copy: $\alpha_{\rm B}$(He II) exceeds $\alpha_{\rm B}$(H II) by 8% at this temperature, so the substitution itself is a small error and the remainder is which mean energy per recombination the various coefficients represent. The channel is worth of order 1% of the HII40 budget.

**Recombination and free-free elsewhere agree well.** MOCASSIN's H$^+$ recombination cooling equals MOCHII's Hui & Gnedin *case-B* coefficient to 0.1% at 8000 K and 1.1% at $10^4$ K (2.346 against $2.343 \times 10^{-25}$, and 2.403 against $2.376 \times 10^{-25}$), and its He$^{++}$ term is 0.97× the case-B He III coefficient, so its recombination cooling is case-B bookkeeping. Its free-free fits correspond to thermally averaged Gaunt factors of 1.259 and 1.272 at $Z = 1$ (8000 and $10^4$ K) and 1.179, 1.191 at $Z = 2$, against 1.252, 1.266 and 1.176, 1.187 from the van Hoof et al. (2014) table MOCHII can select: agreement better than 1%. What is missing is the metals. Free-free is summed over H$^+$, He$^+$ and He$^{++}$ only, with no net-charge term for the metal ions (MOCHII adds one, worth $\sim 2 \times 10^{-4}$ of the H/He term), and there is no metal recombination cooling (0.05% of the CLOUDY budget, Table 1; MOCHII omits it too).

**Collisional ionization is not in the budget.** `collIon` (:1378, Drake & Ulrich 1980) is computed for hydrogen only, explicitly zeroed for every other species (:1485), and enters the ionization ratios (:1490, :1496). It never reaches `coolInt`.

**Dust.** `setDustGasCollHeatCool` (:811–931) is the gas–grain collisional exchange: ions (:831–892) and electrons (:894–924) striking grains at the local grain potential, accumulated as `gasDustColl_g` for the gas and `gasDustColl_d` for the grains. It reaches the budget together with the photoelectric heating at :1253–1256, and both are computed only under `lgDust .and. lgGas .and. lgPhotoelectric` (:368–386), so a dusty model run with the photoelectric switch off has neither. This is the one cooling channel MOCASSIN has and MoCHII does not. Its size at benchmark conditions is negligible: with the standard gas–grain rate (Hollenbach & McKee 1989) at $n_{\rm H}$ = 100 cm$^{-3}$, $T$ = 8000 K, $T_{\rm d}$ = 50 K and Milky Way dust it is $\sim 5 \times 10^{-24}$ erg cm$^{-3}$ s$^{-1}$ against $\sim 4 \times 10^{-20}$ for the lines, of order $10^{-4}$. It scales as $n_{\rm H}^2$ like the lines, so it matters only at planetary-nebula and photodissociation-region densities, and both benchmark models are dust-free. MoCHII routes grain energy to the infrared instead (`Heat_dust`) and carries grain photoelectric heating with grain recombination cooling (`par%grain_pe`).

**Net comparison with MoCHII.** The dominant difference is now the collision data. Both codes suppress the metal line cooling to the local $n_{\rm e}$ in the thermal balance — MOCASSIN by solving the level populations there, MoCHII through the $(T, n_{\rm e})$ table built from its own $n$-level solve (Section 6) — so the $\sim$10% that the suppression removes at $n_{\rm H}$ = 100 cm$^{-3}$ is common to both. What remains is that MOCASSIN's collision strengths are 15–22% higher than the current ones on three of the four largest metal coolants, which adds cooling, while MoCHII carries the current CHIANTI v11 strengths. Neither code's temperature can therefore be attributed to a single data difference. On channels rather than data, MoCHII carries three that MOCASSIN does not — collisional-ionization cooling, metal free-free with the net ion charge, and H-impact fine-structure cooling for the photodissociation-region path (MOCASSIN's $\Upsilon$ tables are electron-impact only) — and MOCASSIN carries one that MoCHII does not, the gas–grain collisional term.

## 7.4 TORUS

TORUS (Harries et al. 2019) is a Monte Carlo radiation transport and hydrodynamics code, and its photoionization loop — the reference the code registers for itself when that loop runs is Haworth & Harries (2012), `citations_mod.f90`:52 — is a full photoionization-equilibrium plus thermal-balance calculation on the same kind of octree MoCHII uses, so it belongs in this comparison. Its provenance runs through both of the codes above. The free-free and photo-heating expressions carry the comments "Kenny equation 22", "Kenny equation 23" and "equation 21 of kenny's" (`photoionAMR_mod.F90`:6465–6467, :7813), that is Wood's paper; the H I collisional-excitation block is MOCASSIN's character for character (below); and two of the ion data blocks say in their own comments that they were tuned to agree with MOCASSIN and with Ercolano's S III data (`ion_mod.F90`:609, :1138).

**Two copies of the module; the MPI one is the production path.** `photoion_mod.F90` (3406 lines) is the serial loop, reached from `physics_mod.F90`:412; `photoionAMR_mod.F90` (11179 lines) is the MPI/AMR loop, reached from `physics_mod.F90`:437 and compiled only under `#ifdef MPI`. The two carry independent copies of `HHeCooling`, `getHeating`, `metalcoolingRate` and `solveIonizationBalance`, and they are not identical: the serial one always adds the gas–grain term and has no dust radiative term (`photoion_mod.F90`:1368), the MPI one makes both conditional. Everything quoted below is the MPI version unless stated.

**Method: Lucy estimator with immediate re-emission.** The packet energy is $\epsilon/\Delta t = L_{\rm ion}/N_{\rm packet}$ (:3626), and the photoionization rate of every ion is the Lucy (1999) path-length estimator, $(\epsilon/\Delta t/V)\sum \ell\sigma_i/h\nu$ (accumulated in `updateGrid`:6691–6692, consumed at :6880); the photoheating uses the same estimator with the weight $\ell\sigma(h\nu - h\nu_{\rm th})/h\nu$ (:6717–6728). When a packet is absorbed it is *immediately re-emitted*: the cell's diffuse emissivity spectrum is assembled once per radiation step from the Lyman continua (:3325), the higher continua (:3326), the H I recombination lines (:3328), the collisionally excited metal lines (:3330) and the dust continuum (:3335), a new frequency is drawn from it (:3372) and the packet continues in a random direction with its energy unchanged (:3374). So TORUS does not let its collisionally excited line photons escape the way MOCASSIN and MoCHII do: they re-enter the transport, and in a photoionized nebula their only sink is dust absorption, so this is how line-cooling energy reaches the infrared. Because the re-emitted packet keeps the absorbed energy and takes only its *shape* from the emissivity, the emergent line luminosities equal the cooling-rate line luminosities only in a converged model, where heating equals cooling by construction of the temperature solve. An on-the-spot option exists: `nodiffuse` zeroes everything above 13.6 eV in the spectrum before the redraw (:3347–3349, `removeLymanSpectrum` :7858–7868).

**The temperature: three modes.** In the equilibrium mode the ionization balance and the temperature alternate in an inner loop per octal until both change by less than 1% (:3908–3927), and `calculateThermalBalanceNew` (:6109–6286) finds the root of $\Gamma_{\rm tot} - \Lambda_{\rm tot}$ by Brent's method, bracketing first on $0.8T$ and $1.2T$ and falling back to $[T_{\rm min}, T_{\rm max}]$ (:6165–6176), then under-relaxing the step by 0.6, or by 0.1 after iteration 10 (:6132–6135). Unlike MOCASSIN and MoCHII, the ionization balance is *not* re-solved at each trial temperature — the call that would do it sits commented out inside the cooling function (`photoion_mod.F90`:1354) — so the coupling is carried by the outer alternation instead. The time-dependent mode integrates $de/dt = \Gamma - \Lambda$ in ten substeps with the same cooling function (`calculateThermalBalanceTimeDep` :6364–6411). A third mode, `quickthermal` (`inputs_modV2.F90`:3948), skips the balance altogether and sets $T = T_{\rm min} + (10^4\ {\rm K} - T_{\rm min})x_{\rm H^+}$ (:5974–5990). Cells the Monte Carlo has undersampled are forced to $T_{\rm min}$ (:6279–6280), which for the default $T_{\rm min} = 3$ K (`inputs_modV2.F90`:65) is a placeholder rather than a physical temperature.

**Budget: seven terms, and the code counts them itself.** `HHeCooling` (:6413–6648) accumulates, in order: free-free (:6465–6467), H I collisional excitation (:6500–6520), $\rm H^+$ recombination cooling (:6571), $\rm He^+$ recombination cooling (:6582), metal line cooling (:6593), the gas–grain collisional term (:6604–6612) and dust radiative cooling (:6620–6628). The routine's own diagnostic branch asserts `nRates /= 7` (:6636), so seven is the intended count.
Two expressions in the same routine are dead. A Hummer (1994) free-free coefficient and a Hummer (1994) $\rm H^+$ recombination-cooling coefficient are evaluated at :6480–6496 and accumulated into `becool`, which is used only by a debug `write` (:6528–6534) and never reaches `coolingRate`. The free-free that is actually used is the "Kenny" form, $1.42\times 10^{-27}(n_{\rm H^+} + n_{\rm He^+})\,n_{\rm e}\,g_{\rm ff}\sqrt{T}$ with $g_{\rm ff} = 1.1 + 0.34\exp[-(5.5 - \log_{10}T)^2/3]$. That $g_{\rm ff}$ is 1.245 at 8000 K and 1.261 at $10^4$ K, against 1.252 and 1.266 from the van Hoof et al. (2014) table MoCHII can select — agreement to 0.6%. The sum runs over $\rm H^+$ and $\rm He^+$ only, both at $Z = 1$: $\rm He^{++}$ is tracked in the ionization balance (`ion_mod.F90`:215) but contributes neither $Z = 2$ free-free nor recombination cooling, and the metals contribute no free-free.

**H I collisional excitation is MOCASSIN's expression.** :6502–6518 carries `ch12 = 2.47e-8`, `ch13 = 1.32e-8`, `ex12 = -0.228`, `ex13 = -0.460`, `th12 = 118338.`, `th13 = 140252.`, the same `hcRyd` per excitation and the same hard cut below 5000 K as `update_mod.f90`:1071–1089, under the same comment "Mathis, Ly alpha, beta". It therefore reproduces the MOCASSIN numbers quoted above exactly: $2.211 \times 10^{-25}$ at 8000 K and $4.140 \times 10^{-24}$ at $10^4$ K, that is 0.97–0.98 of MOCHII's CHIANTI v11 H I low-density fit.

**Recombination cooling: two tabulated coefficients, no cited source.** $n_e n_{H^+} kT \beta_H$ and $n_e n_{He^+} kT \beta_{He}$ with $\sqrt{T}\beta$ read from a 31-point table over $\log_{10} T = 1$–7 for hydrogen and an 18-point table over 1–4.4 for helium, linearly interpolated in $\log_{10} T$ and set to zero above the helium table's end (:6426–6435, :6544–6569). Neither table carries a provenance comment. Numerically the hydrogen one is case-B recombination cooling: $kT\beta_H = 2.360 \times 10^{-25}$ at 8000 K and $2.397 \times 10^{-25}$ at $10^4$ K, which is 1.007 and 1.009 times MOCHII's Hui & Gnedin case-B coefficient and 1.006 and 0.997 times MOCASSIN's. The helium table is a genuinely separate table, not the hydrogen one reused — which is the defect recorded on MOCASSIN's side above — and it gives $2.619 \times 10^{-25}$ at 8000 K and $2.706 \times 10^{-25}$ at $10^4$ K. At $10^4$ K that is 1.13 times MOCASSIN's copied value and 0.70 times MOCHII's $kT\alpha_B(\text{He II})$, so it sits at the low end of the spread but not at MOCASSIN's extreme.

**Dust enters as two terms.** With `decouplegasdust` false, which is the default (`inputs_modV2.F90`:3951), the gas and the grains share one temperature, the grain thermal emission $4\pi\kappa_P(\sigma/\pi)T^4$ is added to the cooling side of the same balance (:6620–6628) and the grain heating rate is added to the heating side (:7818–7823). With it true the two temperatures are separate, the gas–grain collisional exchange $n_{gas} n_{gr} \sigma_{gr} v_p f\, 2k(T_{gas} - T_d)$ becomes a gas cooling term (:6604–6612; `photoion_utils_mod.F90`:1634–1680, with accommodation factor $f = 11.8$ in ionized gas and 0.16 or 1 in neutral gas, :1666–1674), and the grain temperature is solved from radiative equilibrium plus that exchange (:6253–6271). So TORUS carries the gas–grain collisional channel that MOCASSIN has and MOCHII does not, as an option, and it additionally routes grain emission through the gas thermal balance, which no other code here does.

**Level populations are a statistical-equilibrium solve at the local $n_e$.** `solvePops` (`photoion_utils_mod.F90`:923–997) builds the rate matrix with $n_e q$ in both directions plus the $A$-values (:954–974), replaces the first row by the normalization condition (:942–943), solves it by LU decomposition (:986) and normalizes; the collisional rates are $q = 8.63 \times 10^{-6}\,\Upsilon/(g\sqrt{T})$ with the Boltzmann factor from the observed level energies (`getCollisionalRates` :999–1046), $\Upsilon$ linearly interpolated in $T$ and clamped at the table ends with a one-time warning (:1015–1027). So TORUS, like MOCASSIN and like MOCHII's density-suppressed $\Lambda(T, n_e)$ (Section 6), carries collisional de-excitation inside the thermal balance and saturates the far-infrared fine-structure lines above their critical density. Two qualifications. Every caller passes a population array dimensioned `pops(10)` (`photoionAMR_mod.F90`:7712, :7732, :8089; `photoion_utils_mod.F90`:1115), so no ion model may exceed ten levels; the largest in the file is $C$ II with exactly ten. And the recombination-pumping term is dormant here too: `getRecombs` is called (:940) but the line that would put its rates into the right-hand side is commented out and `matrixB(2:n)` is set to zero (:945), so the populations are purely collisional — the same assumption as MOCHII's Tier-2 models.

**Ion set: fifteen line-cooling ions, hardcoded.** There are no data files. `addIons` (`ion_mod.F90`:194–388) builds the registry — H I, H II, He I, He II, He III, then C I–IV, N I–III, O I–III, Ne I–III and S I–IV under `usemetals`, twenty-two ions in all — and `addLevels` (:428–587) and `addTransitions` (:603–1186) fill levels and transitions from `select case` blocks on the species name, the level energies taken from NIST by the file's own statement (:433). Fifteen ions end up with transitions and therefore with cooling: C I–IV, N I–III, O I–III, *Ne* II, *Ne* III and S II–IV. Ne I and S I are in the ionization balance but have neither levels nor lines. *N* I and *O* I are complete — ten transitions each with tabulated $\Upsilon(T)$, the full set for their five levels (:812–847, :914–937) — and *S* IV is present as a two-level ion with $A = 7.74 \times 10^{-3}$ s$^{-1}$ (:1179), which is CMacIonize's Pradhan (1995) value to three figures. So Torus carries neither of the two channels Wood's code loses, and like CMacIonize it has the *S* IV channel MoCHII was missing. There is no chlorine or argon line cooling: argon ions exist only under `xraymetals` and have no level data at all.
The transition energy is the difference of the two observed level energies (`addTransition2` :1266), the stated wavelength is kept separately for line identification only, and a warning fires if the two disagree by more than 5% (:1273–1279). Torus therefore cannot carry MoCHII's D2 defect by construction. Cooling and line output run through the same routine and the same data — `metalcoolingRate`:7742 and `cellLineLuminosity`:7707–7722 are the same expression with the same `solvePops` call — so it cannot carry Wood's cooling/diagnostic split either.

**Collision-strength vintage: mixed, and N I is the modern calculation.** The tables are per transition and per ion, with temperature grids from two points (*C* IV) to sixteen (*S* II, *S* III), and their lineage is visible in the comments: carbon was "freshly mod[ded] … to agree with MOCASSIN" (:609) and the live *S* III block is the fourth attempt, labeled "Matching Barbara Ercolano's S III data as closely as possible" (:1138), with a Tayal & Gupta (1999) block and two others left commented out above it (:1060–1136). Two spot checks against the numbers above. For *O* III the total $\Upsilon(^3P \to {}^1D)$ from :967–991 is 2.119 at 8000 K and 2.184 at $10^4$ K, so 0.965 of CHIANTI v11's 2.262 and 0.957 of Mocassin's v7 file value 2.283 — the same few-percent band all three sit in. For *N* I, however, $\Upsilon(^4S \to {}^2D_{5/2})$ is 0.2338 at 8000 K and 0.2877 at $10^4$ K (:817–819), which is 0.85 of CHIANTI v11's 0.337 and an order of magnitude above Mocassin's 0.027 and CMacIonize's 0.025. Torus's nitrogen is thus on the modern side of the Tayal (2000) versus Tayal (2006) split, and its *N* I cooling and [*N* I] 5200 diagnostic differ from theirs by that factor.

**Defect: a Fortran exponent typo makes an *S* II *A*-value negative.** In the `addTransition2` call for the *S* II $^2P^{\mathrm{o}}_{3/2} \to {}^2P^{\mathrm{o}}_{1/2}$ fine-structure transition (`ion_mod.F90`:1055) the *A*-value argument is written `1.03-6`, which in Fortran is not $1.03 \times 10^{-6}$ but the expression $1.03 - 6 = -4.97$. The transition therefore enters with $A = -4.97$ s$^{-1}$, which exceeds the sum of the three genuine decay rates out of that level (0.537 s$^{-1}$) and makes its total radiative loss rate negative. Reproducing `solvePops` with the file's own five levels, ten transitions and $\Upsilon(T)$ tables gives a *negative* $^2P^{\mathrm{o}}_{3/2}$ population at every temperature and density tried ($-1.08 \times 10^{-8}$ at $T = 8000$ K, $n_e = 100$ cm$^{-3}$, falling to $-2.15 \times 10^{-5}$ at 20 kK and $10^4$ cm$^{-3}$), which the `pops = max(0, ...)` clamp at `photoion_utils_mod.F90`:992 then sets to zero. The consequences are specific. The three [*S* II] lines whose upper level is $^2P^{\mathrm{o}}_{3/2}$ — 4068.6, 10286.7 and 10320.4 Å — emit exactly zero, while the three whose upper level is $^2P^{\mathrm{o}}_{1/2}$ — 4076.4, 10336.3 and 10373.3 Å — come out a factor 3.2 too strong, because the negative level acts as a sink that inflates $^2P^{\mathrm{o}}_{1/2}$ from $7.28 \times 10^{-8}$ to $2.34 \times 10^{-7}$. The intrinsic 4069/4076 ratio, 3.04 on the intended *A*, comes out 0. The *total* *S* II cooling barely notices, because the two errors have opposite sign: 1.005 of the intended value at 8000 K and $n_e = 100$, 1.014 at $n_e = 10^4$, 1.025 at 20 kK and $10^4$. With [*S* II] at 2.16% of the HII40 budget (Table 1) the thermal balance is unaffected at the $10^{-4}$ level; what is destroyed is the [*S* II] auroral diagnostic. The "Negative contribution from" warning writer (`photoionAMR_mod.F90`:7745–7749) exists but tests the summed ion rate, which stays positive.

**Defect: the *N* II $^1S_0$ statistical weight.** `ion_mod.F90`:479 enters the *N* II $^1S_0$ level with $J = 2$, so $g = 5$ instead of 1; every other $^1S_0$ in the file is entered with $J = 0$ (*C* III :456, *O* III :516, *Ne* III :532, *S* III :549). Only the de-excitation rate divides by the upper level's $g$ (`photoion_utils_mod.F90`:1042), so the error is confined to collisional de-excitation out of $^1S_0$; with $n_{\rm crit} \sim 2 \times 10^7$ cm$^{-3}$ for that level it is negligible at nebular densities and wrong by a factor 5 in the de-excitation rate wherever it is not. Mg I carries the same slip on its $^1P^{\rm o}_1$ level (:564), harmlessly, because magnesium has no transitions.

**The X-ray metal set cannot be used as written.** `xraymetals` adds twenty-one Mg, Si, Ar and Fe ions to the registry (`ion_mod.F90`:296–375), but `returnAbundance` (:1317–1345) has no case for $Z = 12$, 14, 18 or 26 and returns `tiny(a)` with a runtime message, and none of the four elements has transition data (magnesium has levels but no transitions; silicon, argon and iron have neither). So the switch contributes no cooling and no opacity of consequence. A latent companion: Mg II declares two levels both named `"2S_1/2"` (:567, :570) and `returnLevel` (:1298–1315) returns the first match, so the second would be unreachable if transitions were ever added.

**What is absent.** There is no collisional ionization anywhere: `solveIonizationBalance` (:6838–6918) forms each stage ratio as (photoionization + charge-exchange ionization) over (radiative + dielectronic recombination + charge-exchange recombination), with no collisional term, so there is no collisional-ionization cooling either. There is no He I or He II collisional-excitation cooling — the metal sum starts at registry entry 5 (:7737), which is He III with no level data, so effectively at *C* I, and helium has no level data at all. There is no He III recombination cooling, no $Z = 2$ free-free, no metal free-free, no metal recombination cooling and no two-photon term (the two-photon routine is commented out, :7775–7799). On the heating side only H I and He I photoionization are counted (:6717–6728, :7812–7817): He II photoionization heating is missing although He$^{++}$ is tracked, and there is no metal photoheating, no grain photoelectric heating and no cosmic-ray heating in the photoionization loop. A separate photodissociation-region module does carry Weingartner & Draine photoelectric heating, cosmic-ray, turbulent, chemical, gas–grain and $H_2$-formation heating (`pdr_utils_mod.F90`:366–800), invoked from the photoionization loop under `pdrcalc` (`photoionAMR_mod.F90`:527, :997), but that is a second thermal balance with its own inventory and is outside this comparison. One point in TORUS's favor on the electron side: metal electrons are always in $n_e$, since `returnNe` (`ion_mod.F90`:1410–1433) sums $Z - N$ over the whole registry, where MoCHII needs `par%metal_ne`.

**The benchmark directory.** A Lexington HII40 model is shipped with the code. `benchmarks/HII_region/` and `benchmarks/HII_regionMPI/` are the same model in the OpenMP and MPI builds: `geometry lexington`, a 1D octree at depth 8, one blackbody source at $T_{\rm eff} = 40000$ K with $R = 18.67\,R_\odot$ (`parameters.dat`), and $n_{\rm H} = 100$ cm$^{-3}$ uniform outside `rinner` (`calcLexington`, `amr_mod.F90`:7557–7612). The default abundances — He 0.1, C $2.2 \times 10^{-4}$, N $4 \times 10^{-5}$, O $3.3 \times 10^{-4}$, Ne $5 \times 10^{-5}$, S $9 \times 10^{-6}$ (`inputs_modV2.F90`:4107–4133) — are exactly the Lexington/Meudon

HII40 composition, so TORUS's photoionization defaults are set by the same benchmark MoCHII is measured against. The output dump (`dumpLexingtonMPI`, `photoionAMR_mod.F90`:7353–7480) writes 500 radial points from 1 to 8 pc: radius, $T_e$, then $\log x$ for H I, He I, $O$ II, $O$ III, $C$ II, $C$ III, $C$ IV, $N$ II, $N$ III, N IV, Ne I, $Ne$ II, $Ne$ III, Ne IV and $T_d$, plus `ne.dat` and an `orates.dat` carrying the cell heating and cooling rates.

The reference file `lexington_benchmark.dat` is committed to the repository, so it is TORUS's own stored regression output rather than a published table, and `comparelex.f90` checks a new run against it in $T_e$ and one ion at a 10% tolerance. It is still usable as a fifth column: weighting the stored profile the way the published benchmark tables do gives $\langle T[N_p N_e]\rangle = 8196$ K, with the $x_{HI} = 0.5$ front at 4.75 pc. Two caveats before that number is used. The dump starts at 1 pc, so the innermost parsec is not in the average; and it is a single stored run at whatever packet and iteration count produced it, not a converged reference in the sense of Section 5. Taken at face value it sits at the top of the 7720–8199 K published range of Table 5, 0.4% above MoCHII's current 8169 K and 0.2% below CLOUDY c25.00's 8210 K (and above MoCHII's zero-density 7894 K). `benchmarks/HII_region/` also holds a `lexington_cloudy.txt` comparison profile. Nothing here was re-run.

**Net comparison with MoCHII.** On channels TORUS is the closest of the four other codes to MoCHII in what it solves and the farthest in what it omits. It has the density-dependent level populations inside the thermal balance, as MoCHII now does through the $(T, n_e)$ table, on transition energies taken from observed levels as MoCHII's rebuilt fits are, and from the same routine that makes its line output; it has the gas–grain collisional term as an option; and it puts grain thermal emission into the gas balance and returns the metal line photons to the transport, neither of which MOCASSIN or MoCHII does. Against that it omits five channels MoCHII carries — collisional-ionization cooling, He III recombination cooling, $Z = 2$ free-free, metal free-free and He II photoheating — caps every ion at ten levels, and cannot be extended past C, N, O, Ne and S without editing Fortran, since its atomic data are `select case` blocks rather than files. On data its collision strengths are a mixture whose $O$ III agrees with CHIANTI v11 to 3.5% and whose $N$ I is an order of magnitude above MOCASSIN's, so the systematic 15–22% excess found above in MOCASSIN's largest metal coolants does not carry over to TORUS unexamined.

## 7.5 Comparison, and one omission common to all five codes

Table 8 collects the result. The negative result of the audit is the last row: He I collisional-excitation line cooling is absent from all five codes, MoCHII included. In the HII40 budget of Table 1 CLOUDY puts $1.943 \times 10^{35}$ erg s$^{-1}$ in it, 0.062% of the total, so at benchmark conditions the omission is harmless; it would matter only where neutral helium is abundant and the gas hot enough to excite it.

Ranking the Wood findings by what they are worth at HII40, using the CLOUDY channel shares of Table 1: the $N$ III collision-strength inconsistency is the largest, a factor 2.1 on 1.89% of the budget; the missing H I collisional excitation is 0.72%; the unread chlorine and argon blocks are 0.093% and 0.006%; the missing He III recombination cooling is 0.001% at this excitation. The N I/O I defect is negligible at HII40 ([$O$ I] is 0.10%, [$N$ I] 0.014%) but it lives precisely in the partly neutral gas, so it matters for the ionization-front edge, for diffuse ionized gas, and for anything PDR-like. The $O$ III $^1S_0$ energy slip does not move the cooling but biases [$O$ III] 4363 and the temperature diagnosed from it by about 2%.

Table 8: Cooling terms of the five Monte Carlo codes. MOCHII's H/He inventory is `src/cooling_mod.f90`:225–258. Ion counts: Wood, CMACIONIZE and TORUS are fixed lists compiled into the source; the MOCASSIN entry is what its data files supply for the HII40 composition (C, N, O, Ne, S) at the default `nstages = 7`, reaching O VI; MOCHII's 28 are its cooling fits, H I plus 27 metal ions over 11 elements, reaching stage IV (V for Ca). MOCHII is a superset of the line and H/He channels except for the last two rows: the gas–grain term, which MOCASSIN carries and TORUS carries as an option; and He I collisional excitation, which no code carries. It now carries the density-dependent level populations in the thermal balance too, through the $(T, n_e)$ table of Section 6.

| Cooling term | Wood | CMACIONIZE | MOCASSIN | TORUS | MOCHII |
|---|---|---|---|---|---|
| metal collisionally excited lines | 12 ions | 13 ions | 26 ions | 15 ions | 28 ions |
| Cl, Ar line cooling | data unread | no | yes | no | yes |
| N I, O I line cooling | defective | yes | yes | yes | yes |
| S IV line cooling | no | yes | yes | yes | yes (added here) |
| H I collisional excitation | no | no | yes | yes | yes |
| collisional-ionization cooling | no | no | no | no | yes |
| He III recombination cooling | no | no | yes | no | yes |
| free-free at $Z = 2$ (He III) | no | no | yes | no | yes |
| free-free from the metal ions | no | no | no | no | yes |
| level populations at the local $n_e$ | yes | yes | yes | yes | yes (table) |
| gas–grain collisional cooling | no | no | yes | optional | no |
| He I collisional excitation | no | no | no | no | no |

# 8. Summary

The MOCHII temperature excess on the Lexington/Meudon benchmarks, which had survived the elimination of charge exchange, reference convergence, spatial resolution, recombination coefficients, collision-strength vintage, band discretization, transport treatment and trace-species coupling, was two defects in the offline construction of the Tier-1 cooling fits: ground-term level populations set by Boltzmann factors, which at $kT \gg \Delta E_{fs}$ is the high-density limit and suppressed the far-infrared fine-structure cooling by 23.9% of the total, and the Burgess–Tully scaling energy used as the physical transition energy in both the excitation rate and the radiated photon, worth a further 10.6%. The diagnosis needed a fixed-state comparison, because in a converged model the temperature rises until the books balance and hides the deficit; the decisive internal evidence was that MOCHII's own $n$-level solve and its own cooling fits disagreed by 28% on an identical state. Rebuilding all 28 fits from the $n_e \to 0$ limit of statistical equilibrium with observed level energies, and adding $S$ IV, moved HII40 inside the published temperature range and closed the photon budget, the outer radius and the line ratios along with it. That zero-density limit still overcooled the low-critical-density lines at $n_H = 100$; suppressing $\Lambda$ to the local $n_e$ with a $(T, n_e)$ table built from the same $n$-level solve — now the default, `par%cooling_model = local_ne` — removes the far-infrared overestimate and raised the converged benchmarks at once to 8211 and 7025 K, matching CLOUDY c25.00 (8210, 6910 K); the zero-density fits are kept as the `low_density` option. Later corrections to the stellar and diffuse radiation field, none of them in the cooling, brought the production figures to their current values: 8169 K at HII40, inside the published 7720–8199 K range and 0.50% below CLOUDY; and 6925 K at HII20, above the published 6402–6749 K range but within 0.22% of CLOUDY c25.00's 6910 K, which is itself 2.4% above that range. The HII20 range is a 1995-era nine-code spread that both current-generation codes exceed, so the verdict there is agreement with CLOUDY, not a shortfall against the published floor.

The same audit applied to the other Monte Carlo photoionization codes shows that MoCHII's inventory is now a superset of all four in the line and H/He channels, with two exceptions. He I collisional-excitation line cooling is absent from Wood's code, from CMACIONIZE, from MOCASSIN, from TORUS and from MoCHII, and is worth 0.062% of the HII40 budget. And MOCASSIN carries a gas–grain collisional term and TORUS carries one as an option, of order $10^{-4}$ of the line cooling at $n_{\rm H} = 100$ cm$^{-3}$. The metal level populations at the local $n_{\rm e}$ that the other four codes solve inside the thermal balance MoCHII now matches through the $(T, n_{\rm e})$ table (Section 6), so that is no longer a difference. It also turned up defects on their side — Wood's N I and O I cooling is zeroed by an overwrite of the collision strengths, its cooling and diagnostic routines disagree by a factor 2.1 on the $N$ III 57.3 $\mu$m collision strength (CHIANTI v11 supports the diagnostic value), its chlorine and argon data are present but never read, its S IV output slot is never assigned, and its $O$ III $^1S_0$ energy is low by 113.7 cm$^{-1}$. CMACIONIZE carries none of those, and the 11% He$^+$ difference between the two is in the recombination *cooling* coefficient, not the recombination rate, which they share exactly. TORUS carries a Fortran exponent typo, `1.03-6` for $1.03 \times 10^{-6}$, that makes one $S$ II $A$-value negative and zeroes three [$S$ II] lines while inflating three others by 3.2, and it enters the $N$ II $^1S_0$ statistical weight as 5 instead of 1.

**Where the numbers come from.** The CLOUDY reference runs and their output files are in `tests/cloudy_c25/` (`hii40_cool.in` for the cooling inventory, `hii{40,20}_c25.in` for the line lists and structure); the benchmark runs after the fix are `tests/g2_hii/hii{40,20}_bench.in` with their `_rates.h5` and `_lines.txt` output; the coefficient files are `data/atomic/cooling_<ion>.txt` and `nlevel_<ion>.txt`, each with its own provenance header; the fitting pipeline is `tools/fitting/{chianti_cooling,fit_cooling,fit_nlevel}.py`; and the benchmark table columns are assembled by `paper/tables/make_wood_tables.py`. The channel-by-channel budget, the defect isolation and the temperature closure were computed by analysis scripts kept in the working session rather than committed, since each of them is a one-off reading of the files listed above.